GENERAL THETA FOUNDATIONS I: ACQUIRED GEOMETRY AND CAUSAL RESOURCE TRANSFER

QIAN QI

Abstract. We derive finite-acquisition risk laws for known prepared positive causal instruments with a finite separating affine family of executable tests. A positive-cone comparison lifts actual acquired overlap to a minorization of the joint physical–observer ensemble. Robust access to a common report region makes its constants independent of the number of retained labels. Together with an invariant calibration correspondence and an average-cost Bellman equation, this constructs one homogeneous observer with at most M total labels and excess predictive risk between $cM^{-2/d}$ and $C(M^{-2/d}+n^{-1})$ against every full-history adaptive acquisition strategy in a protected random-audit task. The same observer works for all horizons and preparations, without artificial erasure, physical reset or an uncounted posterior register. The lower uses actual acquired mass; the upper uses a global legal cover and nonsmooth Bellman curvature. Controlled Gaussian filters and noncommuting quantum instruments verify the same theorem from raw kernels. A weighted acquired-stratum extension, an explicit binary benchmark, an attained calibration–deficiency composition, and finite-precision execution with budget-independent perturbation amplification are proved. All call, clock, program and workspace costs remain distinct.

1. Introduction

A prepared experiment supplies conditional laws of future observations, not an a priori information manifold. Once the legal future tests are fixed, two histories carry the same predictive state precisely when no such test can distinguish them. Finite resources impose a second, different equivalence: an executable observer must select its next experiment, update its retained information and produce its eventual response through the same finite state. A cover of the predictive space need not be realizable by that observer. Nor need a stationary quantizer be calibrated by the law that its own acquisition decisions produce.

The problem considered here is to derive finite-resource decision laws from this actual acquired geometry. We work with a nontrivial but explicit class: known, time-homogeneous positive causal instruments having a finite separating affine family of executable tests. The hypotheses are given before a policy, codebook or invariant law is selected. A density bound measures actual acquired mass; a common grid provides a global legal cover; positive-cone comparison and robust access to a common report component connect that geometry to a realizable recursion. Finite-dimensional closure, whole-base interiority and accessibility are substantive assumptions, not consequences of positivity alone.

Our main result, Theorem 3.2, constructs a single globally calibrated homogeneous observer and proves, for a protected randomly timed predictive audit,

$$cM^{-2/d} \le \mathcal{R}_{\mathrm{hom}}(n,M) - J_*(n) \le C\big(M^{-2/d}+n^{-1}\big). \tag{1}$$

Here $J_*(n)$ optimizes over every legal full-history adaptive acquisition policy in the same experiment. The observer uses at most M total retained labels, including its acquisition control, and the same table works for every horizon n and every admissible preparation. The actual support dimension d

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is derived from executable tests and report pushforwards. It is not the ambient parameter count. The orders match for $n \gtrsim M^{2/d}$ and in the average limit. The theorem does not claim a matching n^{-1} lower term or a complete short-horizon classification.

The mechanism is a lift from acquired mass to joint physical–observer mixing. Interior physical states dominate a uniform positive fraction of every programmed codepoint. Applying the raw instrument to that fraction turns a common component of its acquired law into a common component of the full unnormalized output ensemble, including the next label. Robust access to a small common region extends this one-step lift to informative instruments whose globally acquired rows do not overlap in one step. Crucially, the fraction and block length do not deteriorate as the number of labels grows. This holds for every sufficiently fine local common-grid table, before calibration or optimal control is imposed.

A compact invariant-ensemble correspondence then solves calibration and acquisition control simultaneously. It includes zero-mass labels and tied optimal actions. At its fixed point, the average Bellman equation reduces excess risk to an acquired-mass Jensen loss. Nonsmooth concavity, rather than a globally bounded Hessian, yields the second-order grid rate. The new joint minorization attracts this stationary law from every preparation without inserting an artificial label erasure, resetting the physical experiment or keeping an exact posterior register. It also amplifies per-call numerical ensemble errors by a constant independent of M.

The checkpoint identity and the lower in (1) use the actual acquired law at every scored cut. Conditional projection compares each observer with its own full-history acquisition execution; optimizing afterwards gives the all-controller converse. This avoids comparing a same-controller compression bound with an unrelated unrestricted control optimum. The external audit clock is separately paid and hidden from the observer. Internally exact-deadline tasks have a different state constraint and are not silently identified with this task.

Two structurally different raw experiments verify the same theorem. A controlled finite-state Gaussian filter has a common acquired component at every retained probability. A noncommuting quantum instrument reaches a common component through a positive-probability sequence of quiet reports; arbitrary action-dependent unitaries remain allowed. Section 8 gives an explicit two-label Gaussian filter, its scalar fixed-point equation and a numerical-free mixing certificate. Marked affine acquired strata yield a weighted singular profile with their true branch masses. The common-task extension adds an attained calibration alternative and causal simulation deficiency, while effective compilation separates labels, program constants, workspace, physical calls and the audit clock.

The finite predictive quotient, Bayesian conditional barycenters, acquired-curvature estimate and earlier stationary fixed-point approach are inherited results [8, 9, 10]. They are restated with proofs where they carry this theorem. The new step is the budget-independent acquired-mass lift and its joint use in calibration, finite-acquisition entrance and precision.

2. Prepared experiments and the protected task

2.1. **From executable tests to positive instruments.** Fix a time-homogeneous prepared causal experiment on standard Borel spaces, a finite nonempty action set A, and its legal finite future experiments and bounded readouts. A strategy uses only past reports, actions and independent private randomness. Histories are predictively equivalent when they have the same legal action interface and all executable future-test expectations agree.

We require a finite separating affine test realization. More precisely, the constant test and finitely many executable tests give a Borel vector $u(h) \in \mathbb{R}^d$ whose attained image is Borel and has a Borel realization section. Equality of these coordinates is equivalent to equality of all future tests. The unnormalized expectation of every coordinate test jointly with a next report is affine in the incoming coordinates. Realizable random mixtures form a compact convex base $B \subset \mathbb{R}^d$ with

nonempty interior in this chart. We write

$$C = \{(w, wu) : w \geq 0,\ u \in B\} \subset \mathbb{R}^{d+1}, \qquad e(w, v) = w.$$

This is a closed pointed generating cone, and e is its normalization functional. The associated base norm is

$$\|z\|_B = \inf\{e(z_+) + e(z_-) : z = z_+ - z_-,\ z_\pm \in C\}.$$

On a finite direct sum of label cones we use the sum base norm.

For each action a let λ_a be a reference probability and $L_a(y)$ a Borel, integrable family of positive linear maps on C, with $\bar{L}_a = \int L_a(y)\lambda_a(dy)$ preserving e. Positivity is positivity on this cone; complete positivity is separately verified for quantum instruments. Put

$$L_a(y)(1, u) = (\ell_a(y, u), U_a(y, u)), \qquad F_a(y, u) = U_a(y, u)/\ell_a(y, u). \tag{2}$$

Both ℓ_a and U_a are affine in u. All identities hold for every incoming state off one reference-null set for each action. At zero likelihood the numerator is zero by positivity and bounded tests, and no normalized state is used on that event. The actual acquired row and its integration operator are

$$\mu_{a,u} = (F_a(\cdot, u))_\#(\ell_a(\cdot, u)\lambda_a), \qquad K_a f(u) = \int f(v)\mu_{a,u}(dv). \tag{3}$$

A concrete experiment may first be given on a larger physical state space. Its asserted finite test image and positive maps must then be derived from that experiment, as in Section 7.

Lemma 2.1 (Predictive realization)**.** *Under the preceding hypotheses, the attained test image realizes the predictive quotient as a Borel causal state space. It is minimal among sufficient Borel statistics whose attained images have Borel realization sections. Report and update kernels descend jointly in the finite action coordinate.*

Proof. Equal test coordinates imply equal future tests by the separating realization hypothesis; the converse holds because the coordinates are executable tests. For another sufficient statistic, compose its realization section with $h \mapsto u(h)$. Sufficiency makes this map independent of representatives, proving Borel factorization. The common null sets and the affine instrument identities give Borel descended kernels. Successive integration proves realization of every legal finite continuation. This proves the asserted minimality; it does not assert that an arbitrary measurable quotient is standard Borel. The larger mixture domain B need not be the support of any one acquired law. □

2.2. **Prediction, acquisition and audit.** A fixed bounded executable audit Z has conditional mean u and affine second moment $c(u)$. A response $r \in B$, committed before the audit, has squared loss

$$\mathbb{E}[\|Z - r\|^2 \mid u] = c(u) - 2\langle u, r\rangle + \|r\|^2, \qquad g(u) = c(u) - \|u\|^2. \tag{4}$$

An injective affine probability chart of a categorical or informationally complete measurement gives the same formula on its affine image, with the induced Euclidean norm. Orthogonal components of the actual audit noise remain in c; they are not discarded. Assume the loss is bounded by L_0 on the stated response set.

An infinite unaudited reference execution is used to define costs, not to perform incompatible audits at all times. For an integer $n \geq 1$, an external auditor independently chooses T uniformly from $\{1, \ldots, n\}$, hides T, and requests the response immediately after report T. It then performs the single protected audit and ends the experiment. Before that request the controller sees only the prescribed acquisition reports. The expected risk is the average of the potential single-audit risks at cuts $1, \ldots, n$. For a quantum experiment no intermediate audit is physically inserted.

This protocol uses at most n raw acquisition calls, one initial preparation and one audit, hence a pathwise physical-call budget $N = n + 2$. Its expected raw-report count is $(n + 1)/2$, not n. The auditor pays for its uniform draw and an $\lceil \log_2(n + 1) \rceil$-bit counter. The observer is not given that

counter or an impending-last-report signal. This external clock is recorded separately from observer memory and physical time.

Write X_t for the full-history predictive vector on the reference execution. The full-history comparator may use every legal adaptive action strategy, including its own time and workspace; its risk is

$$J_*(n) = \inf_{\pi} \frac{1}{n} \sum_{t=1}^{n} \mathbb{E}_\pi g(X_t). \tag{5}$$

A checkpoint has the same unrestricted acquisition history and, when requested, encodes it into M labels with one label-only decoder. Its acquisition workspace is not identified with M.

A homogeneous finite observer has a single set S of at most M labels, an initial label, a fixed action distribution p_s at each label, a Borel update distribution $\gamma(j \mid s, a, y)$ and a fixed readout r_s. Its only retained dynamic variable is its label. A fresh action/update coin is independent of the physical state given that label; a retained seed would count as state. Neither a history tape, a real posterior register, a phase counter nor the previous report is available at the next call. The external request reads the existing label and ends the run. Put $\mathcal{R}_{\mathrm{hom}}(n, M) = \inf R_n$ over these programs. Their tables may depend on the known experiment and on M; the controller constructed below works for every n and every initial preparation. Known real table entries are counted as program data, not as costless bit strings.

3. Acquired accessibility and finite resources

The next certificates precede the choice of an observer. In particular, they do not assume that a quantized observer is mixing or that its centroids are calibrated.

Definition 3.1 (Accessible positive instruments)**.** In Section 2, let $K \subset \operatorname{int} D \subset B$ be compact convex sets, full-dimensional in the executable test chart, with $K + B_r \subset D$ for some $r > 0$. Impose the following conditions, with constants independent of any program or budget.

Raw interior image. Off a common reference-null set for each action,

$$L_a(y)C \subset C_K := \{w(1, u) : w \geq 0,\ u \in K\}. \tag{6}$$

This is required on the entire physical cone C, not just on normalized states in D. Likelihoods are positive on D. The integrable finite-dimensional positive maps and affine closure are those of Section 2.

Acquired regularity. On D, every actual row is weakly continuous and

$$\mu_{a,u} \leq \Lambda \operatorname{vol}_d|_K, \qquad \mathrm{TV}(\ell_a(\cdot, u)\lambda_a, \ell_a(\cdot, v)\lambda_a) \leq L_{\mathrm{rep}}\|u - v\|. \tag{7}$$

Use common Borel grid cells C_j whose compact convex closures in K have diameter at most $C'_K M^{-1/d}$, with at most M nonempty cells. Boundaries have zero mass under every row. Convex K and (7) supply these grids; no stationary law is supplied.

Local actual overlap. For a closed $G \subset K$, a probability ζ on K and $\epsilon > 0$,

$$\mu_{a,q} \geq \epsilon\zeta \quad (q \in G,\ a \in A). \tag{8}$$

Either $G = K$, in which case put $k = 0$ and $\theta = 1$, or the following robust accessibility certificate holds. There are Borel report sets E_a, a nonnegative L_V-Lipschitz function V on K, constants $R, v > 0$, $0 \leq \lambda < 1$, $c \geq 0$, $r_G > 0$, an integer $k \geq 1$ and a mesh allowance $s_0 > 0$, such that

$$V \leq R, \qquad \{u \in K : V(u) \leq r_G\} \subset G, \qquad \inf_{a \in A, x \in K} \int_{E_a} \ell_a(y, x)\, d\lambda_a(y) \geq v, \tag{9}$$

$$V(F_a(y, q)) \leq \lambda V(q) + c \quad (q \in K,\ y \in E_a), \tag{10}$$

$$\lambda^k R + (c + L_V s_0) \sum_{i=0}^{k-1} \lambda^i \leq r_G. \tag{11}$$

The drift assertion is off the common null sets. Set $\theta = v^k$. When $k = 0$ the mesh allowance is unrestricted. Only the specified reports must decrease V; other reports and optimal action choices need not be contractive.

Compact containment of K in the interior of the physical base supplies $\kappa > 0$ such that

$$(1, x) - \kappa(1, q) \in C \quad (x, q \in K). \tag{12}$$

We take $\kappa < 1$, and set $b = k + 1$ and $\gamma = \kappa\epsilon\theta \in (0, 1)$. These quantities do not depend on M. The acquired overlap for the full-history process, derived rather than assumed separately, is

$$\mathrm{Law}_u^\pi(X_{k+2}) \geq \epsilon\theta\zeta \quad (u \in D,\ \pi \text{ any legal } (k+2)\text{-call feedback policy}). \tag{13}$$

Theorem 3.2 (Erasure-free acquired-geometry transfer)**.** *Under Definition 3.1 and the prepared executable-task hypotheses, there are a continuous concave average-cost bias h on D and a scalar ρ such that*

$$\rho + h(u) = g(u) + \min_{a \in A} K_a h(u). \tag{14}$$

Moreover $h + \|\cdot\|^2$ is concave, and, with $H = \mathrm{osc}_D\, h$,

$$H \leq \frac{(k+2)\,\mathrm{osc}_D\, g}{\epsilon\theta}, \qquad \mathrm{Lip}_K\, h \leq H/r, \qquad |J_*(n) - \rho| \leq C_0/n. \tag{15}$$

There is a fixed M_0 such that for every $M \geq M_0$ one homogeneous observer γ_M uses at most M labels and, simultaneously for all $n \geq 1$ and preparations $u_0 \in D$, has the following properties.

Its action rows and readouts are calibrated by its unique actual invariant ensemble. It uses a common report grid and has no erasure label, physical reset, retained phase counter or exact posterior register. Its positive ensemble map P_M satisfies, on the direct-sum cone C_K^M,

$$P_M^b Z \geq_{C_K^M} \gamma e_{\mathrm{tot}}(Z) Z_M^\circ, \qquad e_{\mathrm{tot}}(Z_M^\circ) = 1, \tag{16}$$

for an explicitly constructed ensemble Z_M°. Consequently

$$\rho + cM^{-2/d} \leq R_\infty(\gamma_M) \leq \rho + CM^{-2/d}, \tag{17}$$

$$R_n(\gamma_M, u_0) \leq \rho + CM^{-2/d} + \frac{2L_0 b}{n\gamma}. \tag{18}$$

The same observer and table work for every n and every initial label. Its ensemble distance from stationarity after the first acquisition is at most $2(1-\gamma)^{\lfloor (t-1)/b \rfloor}$ in the sum base norm associated with C_K.

Against the unrestricted full-history acquisition comparator in the same protected task,

$$cM^{-2/d} \leq \mathcal{R}_{\mathrm{hom}}(n, M) - J_*(n) \leq C\big(M^{-2/d} + n^{-1}\big). \tag{19}$$

The lower covers every legal randomized Borel observer with at most M retained labels, including time-dependent tables. The upper does not grant that larger architecture to the constructed observer. Thus

$$\inf_{\gamma'} \limsup_{n \to \infty} R_n(\gamma', u_0) - \rho \asymp M^{-2/d}, \tag{20}$$

and the finite-acquisition orders match already when $n \geq c_* M^{2/d}$ for any fixed $c_* > 0$. Constants may depend on c_* and the fixed certificates, but not on n, M.

The finite-n conclusion does not claim an unavoidable n^{-1} lower term. In the complementary short-acquisition regime the displayed bounds need not match. The physical-call budget is $N = n+2$ for the hidden uniform audit, not n independent samples or an internally clocked exact deadline. The cone minorization is a proof decomposition of the executed kernel, not a new resetting action. Both Z_M° and the invariant ensemble retain actual subprobability masses.

Corollary 3.3 (Checkpoint and all-observer lower)**.** *For* $\nu_n^\pi = n^{-1}\sum_{t=1}^n \mathrm{Law}_\pi(X_t)$ *and* $B_n^\pi = n^{-1}\sum_{t=1}^n \mathbb{E}_\pi g(X_t)$*,*

$$\mathcal{R}_{\mathrm{cp}}(n,M) = \inf_\pi\{B_n^\pi + q_M(\nu_n^\pi)\}, \qquad cM^{-2/d} \le \mathcal{R}_{\mathrm{cp}}(n,M) - J_*(n) \le CM^{-2/d}. \tag{21}$$

Here $q_M(\nu) = \inf_{r_1,\dots,r_M}\int \min_j \|x-r_j\|^2\nu(dx)$*, with readouts in* B*. The decoder has no free time argument. The lower in* (19) *follows for every* n *without subtracting a Bellman boundary term.*

4. Average-cost bias and acquired curvature

Lemma 4.1 (Actual feedback-uniform entrance)**.** *Definition 3.1 implies* (13)*.*

Proof. After the first report the exact predictive state is in K. When $k>0$, require the next k reports to lie in E_a for their actual chosen actions. Successive conditional integration gives mass at least v^k, uniformly over every feedback policy. The exact recursion satisfies the drift bound with mesh error zero, so its state is in G. Its next acquired row dominates $\epsilon\zeta$. When $k=0$, that next row has this overlap directly. All $k+2$ calls occur in the actual experiment; the restricted report words are only a subprobability witness. No policy is selected after observing a future report. □

For the following inherited Bellman argument set $b_* = k+2$ and $\epsilon_* = \epsilon\theta$.

Lemma 4.2 (Block span contraction)**.** *Define* $\mathcal{T}f = g + \min_a K_a f$ *on* $C(D)$*. Under* (13)*,*

$$\mathrm{osc}_D(\mathcal{T}^{b_*}f - \mathcal{T}^{b_*}q) \le (1-\epsilon_*)\,\mathrm{osc}_D(f-q).$$

There is a solution of (14)*, unique modulo constants in* h*, with the bounds in* (15)*. It is strongly concave in the stated task chart.*

Proof. For a fixed starting state, compare a b_*-step policy minimizing the terminal-f objective with one minimizing the terminal-q objective. Finite actions and weak continuity give Borel selectors by finite backward induction; arbitrarily minimizing policies give the same inequalities. Running costs cancel when the same policy is used in this comparison. Thus their difference is bounded above and below by terminal expectations of $f-q$ under legal policies. Every such terminal law is $\epsilon_*\zeta + (1-\epsilon_*)\widetilde{\mu}$. Therefore every value of $\mathcal{T}^{b_*}f - \mathcal{T}^{b_*}q$ lies between

$$\epsilon_*\zeta(f-q) + (1-\epsilon_*)\inf_D(f-q) \quad\text{and}\quad \epsilon_*\zeta(f-q) + (1-\epsilon_*)\sup_D(f-q).$$

This proves the contraction, including feedback policies with different state-dependent minimizing actions.

The quotient $C(D)/\mathbb{R}$ with oscillation norm is complete. The translation-equivariant $\mathcal{T}$ induces a map whose b_*th iterate is a strict contraction. Its unique fixed class is also fixed by $\mathcal{T}$, since $\mathcal{T}$ commutes with its iterate. Choose its representative h to vanish at a fixed point of D. Then $\mathcal{T}h = h + \rho$. Since $\mathrm{osc}_D(\mathcal{T}^{b_*}0) \le b_*\,\mathrm{osc}_D\, g$, the contraction gives $H \le b_*\,\mathrm{osc}_D\, g/\epsilon_*$. This is the usual span-contraction argument, applied here to feedback-uniform acquired laws rather than to a latent transition alone [3, 6].

If f is concave, its perspective $wf(v/w)$ is concave on the positive cone of its domain. Hence $K_a f$ is concave by (2). The minimum of finitely many concave functions is concave. Iteration from zero and uniform convergence modulo constants show that h is concave. Equation (14) gives $h - g = \min_a K_a h - \rho$ concave; since $g = c - \|\cdot\|^2$ with affine c, this proves strong concavity without a Hessian assumption.

For a concave function on D, extending the line through two points of K by length r at either end bounds each intermediate slope in absolute value by H/r. Thus h is H/r-Lipschitz on K, including nonsmooth switching surfaces. Bellman minimizers in (14) exist and have a Borel selection. Summing the Bellman inequality along any full-history strategy gives a lower average cost $\rho - H/n$ at cuts $0,\dots,n-1$; a minimizing selector gives an upper $\rho + H/n$. Shifting these cuts to $1,\dots,n$

changes the average by at most $\operatorname{osc}_D g/n$. This proves the asserted bound for $J_*(n)$ and full-history average optimality of ρ. □

Lemma 4.3 (Boundary-valid acquired-curvature estimate)**.** *Let v be concave and L-Lipschitz on compact convex $K \subset \mathbb{R}^d$, and let μ be a probability with density at most Λ. A single grid partition $\mathcal{P}_m$ of K with at most m cells satisfies*

$$\sum_{C\in\mathcal{P}_m} \mu(C) v\left(\frac{\int_C x\, d\mu}{\mu(C)}\right) - \int v\, d\mu \le C_{K,d}\Lambda L m^{-2/d}. \tag{22}$$

Zero-mass cells are omitted. The grid is independent of v, μ; the centroids are not.

Proof. This is the acquired-mass curvature mechanism of [9], recalled to make the stationary proof self-contained. The convex function $-v$ extends to a convex L-Lipschitz function f on $\mathbb{R}^d$: take the supremum of its supporting planes at interior points of K. Interior slopes have norm at most L, and continuity gives equality on the boundary. Its distributional Laplacian $\kappa = \Delta f$ is a positive locally finite measure.

For a cube C of side s and center z, mollify f and subtract its tangent plane at z, obtaining a nonnegative convex w with $w(z) = 0$. On a ball of radius $R = 2\sqrt{d}\, s$, submean estimates bound w on C by a fixed multiple of the average of w on the surrounding sphere. Convex radial slopes give $w(z + R\theta) \le R\partial_R w(z + R\theta)$. Integration and the divergence theorem therefore yield

$$\sup_C w \le C_d R^{2-d} \int_{B_R(z)} \Delta f.$$

For $d = 1$ the sphere consists of the two endpoints and the same calculation applies. Tangent contributions cancel at the conditional centroid, and its value of w is nonnegative. Passing through mollification with a slightly enlarged ball proves

$$\int_C f\, d\mu - \mu(C) f(\mathbb{E}[X \mid C]) \le C_d s^{2-d} \mu(C) \kappa(\overline{B}_{3\sqrt{d}s}(z)).$$

A smooth cutoff on a fixed neighborhood of K bounds its curvature mass by $\int \chi\, d\kappa = -\int \nabla\chi \cdot \nabla f \le L \int |\nabla\chi|$. The enlarged balls have bounded overlap and $\mu(C) \le \Lambda s^d$. Summation gives $C_{K,d}\Lambda L s^2$. Divide a bounding cube into $\lfloor m^{1/d} \rfloor$ parts per coordinate, intersect with K, and discard empty cells. This proves (22), with the case $m = 1$ absorbed by its fixed constant. No positive distance of the acquired support from the boundary of K is used. □

Lemma 4.4 (Actual-law small-ball lower)**.** *Every mixture of the acquired rows in Definition 3.1 satisfies*

$$q_M(\mu) \ge c_d \Lambda^{-2/d} M^{-2/d}, \qquad c_d = \tfrac{1}{2}(2v_d)^{-2/d}, \tag{23}$$

where v_d is the Euclidean unit-ball volume.

Proof. The density bound survives mixing. The union of radius-r_0 balls about M readouts has probability at most $\Lambda M v_d r_0^d$. For $r_0 = (2\Lambda M v_d)^{-1/d}$ at least half the actual mass lies outside. Randomized assignment cannot improve the pointwise minimum over its finite readouts. This is a bound on an acquired law, not on a formally reachable set. □

5. Lifting acquired mass to an executed observer

The full-history process and a finite observer need not have the same mixing properties. The following lift is the step that connects them.

Lemma 5.1 (Positive-cone lift of a local acquired component)**.** *Fix a common partition $\{C_j\}_{j=1}^m$ and representatives $q_s \in \overline{C}_s \subset K$, with arbitrary action distributions p_s. Send report y under action a to the unique cell containing $F_a(y, q_s)$. Define*

$$(Z_m^\circ)_j = \int_{C_j} (1, x)\, \zeta(dx).$$

Then $Z_m^\circ \in C_K^m$ is normalized. For every nonnegative ensemble supported on labels with $q_s \in G$, its next ensemble satisfies

$$PZ \geq_{C_K^m} \kappa\epsilon e_{\mathrm{tot}}(Z) Z_m^\circ. \tag{24}$$

No calibration of the input ensemble is needed.

Proof. For completeness, choose $a > 0$ with $K + B_a \subset B$ and let D_K be the diameter of K. Taking $\kappa > 0$ sufficiently small that $\kappa D_K/(1-\kappa) \leq a$ gives $(x - \kappa q)/(1-\kappa) \in B$, proving (12). For a positive-mass input row $z_s = w_s(1, x_s)$ with $x_s \in K$, write

$$z_s = \kappa w_s(1, q_s) + z_s^{\mathrm{rem}}, \qquad z_s^{\mathrm{rem}} \in C.$$

The remainder need not lie in C_K. Precisely for this reason (6) was imposed on the whole physical cone: its image under every report instrument does lie in C_K. The contribution to output label j is therefore at least, in the C_K order,

$$\kappa w_s \sum_a p_{sa} \int_{\{F_a(y,q_s) \in C_j\}} L_a(y)(1, q_s)\, d\lambda_a(y) = \kappa w_s \sum_a p_{sa} \int_{C_j} (1, x)\, \mu_{a,q_s}(dx).$$

When $q_s \in G$, subtract $\kappa\epsilon w_s (Z_m^\circ)_j$. The remaining measure is positive and supported on K, so its barycentric integral is again in C_K. Sum over the incoming labels. Zero-mass rows contribute zero. The integral identity uses the actual likelihood in the pushforward, not the volume of an image. Grid boundaries have zero ζ mass because $\epsilon\zeta \leq \mu_{a,q}$ for any $q \in G$. □

Lemma 5.2 (Uniform mixing of every local common-grid table)**.** *Under Definition 3.1, take any such table with cell diameter at most s_0 when $k > 0$. Regardless of its action rows and without assuming stationary calibration, its ensemble map satisfies* (16)*, with m in place of M and $b = k+1$, $\gamma = \kappa\epsilon\theta$. Hence it has a unique invariant normalized C_K ensemble and contracts equal-mass differences by $1 - \gamma$ every b calls. These constants are independent of the number of cells.*

Proof. If $k = 0$, every representative lies in G, so Lemma 5.1 proves the assertion. Otherwise consider the subexecution on which the first k reports belong to E_{A_t}. Its actual conditional probability at each step is at least v, whatever the incoming physical state in K, the label, the chosen action and the earlier reports. The total subensemble consequently has mass at least $v^k e_{\mathrm{tot}}(Z)$; this statement follows by successive conditional integration, not independence of the reports.

On this subexecution the programmed representatives obey

$$V(q_{S_{t+1}}) \leq \lambda V(q_{S_t}) + c + L_V s_0.$$

Indeed $F_{A_t}(Y_{t+1}, q_{S_t})$ and $q_{S_{t+1}}$ belong to the closure of the same cell. Equation (11) puts the final representative in G. The actual physical ensemble, which may be quite different from the programmed representative, remains in C_K by (6). Apply Lemma 5.1 to this subensemble on the next report. The complementary executed subensemble is also positive. Thus the full $(k+1)$-step map dominates $\kappa\epsilon v^k e_{\mathrm{tot}}(Z) Z_m^\circ$.

On C_K^m write

$$P^b = \gamma\mathcal{S} + (1-\gamma)Q, \qquad \mathcal{S}Z = e_{\mathrm{tot}}(Z) Z_m^\circ.$$

The residual Q is positive and mass preserving. Such maps contract the base norm, while $\mathcal{S}$ vanishes on zero-total-mass differences. This proves the contraction. The normalized ensemble base is

compact and complete. The contraction gives the unique fixed point of P^b; applying P to it gives another fixed point, so it is also fixed by P. The geometric estimate follows. This is convergence of represented executable-test ensembles, not total-variation convergence of an arbitrary unrepresented microscopic state or of an infinite transcript. □

Lemma 5.3 (Global calibration without erasure)**.** *For each permitted common grid there is one homogeneous table of the preceding form whose invariant positive-mass label means equal its programmed representatives, and whose action rows are supported on* $\arg\min_a K_a h(q_s)$*. The table is defined at all labels, including zero-mass invariant labels, and uses exactly the grid labels.*

Proof. Let $D_s = \overline{C}_s \subset K$ be the compact convex cell closures. Use the compact convex ensemble base

$$\mathcal{E} = \{(z_s) : z_s \in C_{D_s},\ \sum_s e(z_s) = 1\}.$$

At positive mass put $z_s = w_s(1, q_s)$ and allow every probability row supported on the finite minimizing set of $K_a h(q_s)$. At zero mass choose any $q_s \in D_s$ and any action row; its contribution is zero. For each allowed choice define

$$\mathcal{F}_j(z,p) = \sum_{s,a} p_{sa} \int \mathbf{1}_{\{F_a(y,q_s)\in C_j\}} L_a(y) z_s \, d\lambda_a(y). \tag{25}$$

For positive rows the integral is a mass-weighted average of $(1, x)$ with $x \in C_j$. Thus $\mathcal{F}_j \in C_{D_j}$ and total mass is one. Its values are nonempty, convex and compact: for fixed z the map is linear in the product of the minimizing-action simplices; arbitrary zero-row choices have no effect.

The graph is closed. Along a convergent sequence, a positive limiting mass gives convergent normalized representatives and, after taking a subsequence, convergent action probabilities supported on limiting minimizers because $K_a h$ is continuous. The normalized positive instrument is continuous in its input off the common null set. The boundary preimage has zero reference mass, since the acquired row gives the boundary zero mass and its likelihood is positive. Dominated convergence, using the integrable operator norm, handles the cell indicator integrals. At a zero limiting mass every row contribution is bounded by a constant times that mass, uniformly in its allowed representatives. No normalization at zero is used.

The finite-dimensional Kakutani theorem [4] gives $z^* \in \mathcal{F}(z^*)$. Fix one witnessing action selection and choose representatives in their own cell closures for zero-mass labels. The equations are precisely the invariant equations of the executed positive instrument. Its invariant means are therefore the positive-mass q_s themselves, and it is realizable by the permitted prepared mixtures. Lemma 5.2 gives uniqueness and attraction for this fixed table, including from a start not calibrated by z^*. The proof does not assert convergence of an iteration that repeatedly recomputes the nonlinear fixed-point correspondence. □

Proposition 5.4 (Risk of the calibrated table)**.** *The observer of Lemma 5.3 satisfies* (17) *and* (18)*.*

Proof. At its actual invariant ensemble put $Q = q_S$, choose A by the programmed minimizing row, and set $W = F_A(Y, Q)$. The actual law $\mu = \sum_{s,a} w_s p_{sa} \mu_{a,q_s}$ has density at most Λ. Its output mean Q' is the conditional centroid of W in the common cell. Sum the Bellman equation over the calibrated invariant ensemble. Stationarity cancels the incoming and outgoing bias terms and yields

$$R_\infty - \rho = \mathbb{E}h(Q') - \mathbb{E}h(W) = \mathcal{J}_{\mathcal{P}_m}(h, \mu). \tag{26}$$

The curvature estimate in Lemma 4.3 bounds this by $Cm^{-2/d}$. Strong concavity bounds it below by $\mathbb{E}\|W - Q'\|^2 \geq q_m(\mu) \geq cm^{-2/d}$. Choose a grid obtained by splitting a fixed bounding cube with $\lfloor M^{1/d} \rfloor$ subdivisions per side. Its number of cells is at most M, and its mesh and curvature bound are $O(M^{-1/d})$ and $O(M^{-2/d})$, even if some cells are empty. The lower with M follows from the actual number being at most M.

Every initial preparation in D is sent to K after one acquired report. Thereafter Lemma 5.2 bounds the ensemble distance by $2(1-\gamma)^{\lfloor (t-1)/b \rfloor}$. For the fixed readouts, expected audit loss is affine in the unnormalized ensemble and bounded by L_0 on its normalized base. Summing the resulting expectation differences over $t = 1, \dots, n$ costs at most $2L_0 b/(n\gamma)$. This proves (18). Calibration is needed at the invariant law only; no claim of exact calibration during entrance has entered the argument. □

6. The lower bound against every legal observer

Lemma 6.1 (Controlled conditional barycenters)**.** *For any legal finite-state observer let* $Q_t = \mathbb{E}[X_t \mid S_t]$ *and set* $W_{t+1} = F_{A_t}(Y_{t+1}, Q_t)$*. Then*

$$\mathbb{E}[X_{t+1} \mid S_t, A_t, Y_{t+1}] = W_{t+1}, \qquad Q_{t+1} = \mathbb{E}[W_{t+1} \mid S_{t+1}].$$

The identities hold on positive-mass events for randomized Borel updates, also for time-dependent tables.

Proof. The fresh action coin is independent of the physical state conditionally on the label. Affinity of both the unnormalized test numerator and the report likelihood gives conditional integrals $U_a(y, Q_t)$ and $\ell_a(y, Q_t)$. Division at positive likelihood, or equivalently integration against bounded report functions, proves the first identity without conditioning a continuous report on a positive-probability singleton. Fresh update randomness conveys no additional physical information given (S_t, A_t, Y_{t+1}). The tower property proves the second. The full history includes prior private coins, not a future audit. This is the likelihood-reweighted identity of [8, 9], rather than interchange of a nonlinear Bayes update and an unreweighted expectation. □

Proposition 6.2 (Bellman and checkpoint lower bounds)**.** *Define*

$$\Delta_t = \mathbb{E}\big[K_{A_t} h(Q_t) - \min_a K_a h(Q_t)\big], \qquad J_{t+1} = \mathbb{E}h(Q_{t+1}) - \mathbb{E}h(W_{t+1}).$$

For every declared observer,

$$\mathbb{E}g(Q_t) - \rho = \mathbb{E}h(Q_t) - \mathbb{E}h(Q_{t+1}) + \Delta_t + J_{t+1}, \qquad \Delta_t \geq 0, \quad J_{t+1} \geq cM^{-2/d}. \tag{27}$$

In particular

$$R_n \geq \rho + cM^{-2/d} - C/n. \tag{28}$$

For comparison with $J_*(n)$ *the stronger bound* $R_n \geq J_*(n) + cM^{-2/d}$ *holds, as does Corollary 3.3.*

Proof. Substitute (14) at Q_t and add and subtract $\mathbb{E}h(Q_{t+1})$. Lemma 6.1 proves the identity. The law of W_{t+1} is an actual mixture of acquired rows with density at most Λ, even if Q_t is not on the exact filter's support. Strong concavity gives

$$J_{t+1} \geq \mathbb{E}\|W_{t+1} - Q_{t+1}\|^2 \geq q_M(\mathrm{Law}(W_{t+1})) \geq cM^{-2/d}.$$

Sum from $t = 0$ to $n-1$, bound the bias telescope by H, and shift the bounded stage costs to cuts $1, \dots, n$. Any actual label readout satisfies $\mathbb{E}\|Z_t - r_{S_t}\|^2 = \mathbb{E}g(Q_t) + \mathbb{E}\|Q_t - r_{S_t}\|^2$, which proves (28).

The exact finite-acquisition comparison is obtained before discarding information. Fix any unrestricted acquisition policy π, including any causal private coins. Conditional projection at the hidden uniform cut gives

$$R_n = B_n^\pi + \mathbb{E}\|X_T - r_J\|^2.$$

For a checkpoint, minimization over its M label-only readouts is exactly $q_M(\nu_n^\pi)$. Pointwise nearest assignment gives the upper; randomized assignment cannot improve the minimum distance to a fixed finite codebook. Compactness of B permits taking the codebook infimum, and an approximate codebook would suffice without attainment.

For every policy, each positive-time X_t has an actual row-mixture density bounded by Λ. This includes adaptive action selection and the actual preparation at the first call. Thus $q_M(\nu_n^\pi) \geq cM^{-2/d}$ uniformly over π. A common grid with representatives in the cell closures gives the uniform upper $CM^{-2/d}$. Taking the infimum over the same policy class proves (21).

Finally the executed finite observer is one of those acquisition policies, and its existing label is a permissible checkpoint encoder of the complete acquisition history and private randomization. Its fixed readouts have at most M reproduction points. The same projection and small-ball argument therefore proves $R_n \geq B_n^\pi + cM^{-2/d} \geq J_*(n) + cM^{-2/d}$ directly. If the enlarged lower-bound class also uses time-dependent readouts, apply the projection and the uniform small-ball lower separately at each cut before averaging; there are still at most M readouts at that cut. Thus no free-time decoder is inserted into the checkpoint identity. This does not subtract two separately optimized policy baselines. Repeated readouts can only reduce the effective codebook. Together with Proposition 5.4 and (15), it completes Theorem 3.2. □

7. Two raw experiment realizations

The general theorem was proved without using either model below. The finite test closures, acquired laws and accessibility certificates are now derived from their physical kernels. Several density calculations are inherited from [8, 9, 10]; they are included to expose every hypothesis of the new joint-ensemble lift.

Theorem 7.1 (Controlled Gaussian observations)**.** *Let the physical state have $q \geq 2$ values. Let P be invertible and column stochastic with $P_{ij} \geq a_0 > 0$. An action chooses one of finitely many Gaussian sensors: conditional on pre-transition state i, its report is $N(m_i^a, \Sigma_a)$ on $\mathbb{R}^{q-1}$, with Σ_a positive definite and full-rank likelihood-difference matrix. After the report the state transitions by P. A protected categorical audit of the current state is scored by squared probability loss. Then Theorem 3.2 holds with $d = q - 1$; the common-cell observer needs no erasure mechanism and has an entrance term C/n uniform in M. The comparison permits every full-history adaptive sensor choice.*

Proof. The physical probability vector u is separated by the executable categorical audit. Every further finite instrument word has a linear expectation in u; hence its coordinates are exactly the predictive quotient, with the mixture simplex as base and a Borel preparation section. The raw instrument, written relative to Lebesgue measure, is

$$L_a(y)u = P\,\mathrm{diag}(f_1^a(y), \ldots, f_q^a(y))u.$$

Replacing Lebesgue measure by any positive normalized mixture of these Gaussian densities gives the reference probability formulation without changing this instrument. It is positive and integrable. The likelihood is $\sum_i u_i f_i^a(y)$, and the normalized output is P times the ordinary Bayes probability. The task has $g(u) = 1 - \|u\|^2$ on the simplex hyperplane, with its actual audit sampling noise retained.

Take $K = P\Delta_{q-1}$ and $D = \{u \in \Delta_{q-1} : u_i \geq a_0/2\}$. These have the required relative buffer. Every positive physical input, even a boundary preparation outside D, has normalized output in K; this verifies (6) on the entire physical cone. The likelihood is positive on D and its total-variation dependence on u is Lipschitz. Dominated integration gives weak continuity. For $x, q' \in K$, $x_i \geq a_0$ and $q_i' \leq 1$, so (12) holds, for example, with $\kappa = a_0/2$.

Before applying P, write $z_i = \log(w_i/w_q)$ for the Bayes vector w. Then

$$z_i = \log(u_i/u_q) + (m_i^a - m_q^a)^T\Sigma_a^{-1}y - \tfrac{1}{2}[(m_i^a)^T\Sigma_a^{-1}m_i^a - (m_q^a)^T\Sigma_a^{-1}m_q^a].$$

Under the actual report law this is a finite mixture of nonsingular Gaussians. The compact input domain and finite sensor set give a uniform bound $C\exp(-c\|z\|^2)$ on their density. Constants include a positive determinant lower bound for the likelihood-difference matrices and bounds on the

covariance eigenvalues. The inverse logistic Jacobian is a fixed multiple of $(\prod_i w_i)^{-1} \le C\exp(C\|z\|)$. Gaussian tails dominate that factor throughout the simplex, including near every face. The invertible affine map P supplies the density upper on K.

On a fixed full-dimensional compact interior simplex W_*, the same Gaussian mixtures have a uniform strictly positive density lower. Thus $\mu_{a,u} \ge \epsilon\zeta$ on PW_*, with ζ its normalized relative volume, uniformly for $u \in D$ and every sensor. We may set $G = K$, $k = 0$. This is an actual mass bound; no tail or unsuccessful acquisition is discarded. Hyperplane grids supply the boundary-free cells. Lemma 5.1 now yields a one-step joint ensemble minorization for every local common-grid observer. Neither contraction of the sensor selector nor contraction of each posterior path is assumed. □

Theorem 7.2 (Noncommuting controlled quantum instruments)**.** *Let $D_q \ge 2$, $0 < \beta, \lambda < 1$, and let U_a be any finite nonempty family of unitaries. On the unit ball of traceless Hermitian matrices, with normalized Hilbert–Schmidt volume, set*

$$E_H = I + \beta H, \quad K_H = E_H^{1/2}, \quad \tau = I/D_q,$$

$$L_a(H)\sigma = \lambda U_a K_H \sigma K_H U_a^* + (1-\lambda)\operatorname{tr}(E_H\sigma)\tau. \tag{29}$$

Choose the action before the report H, and use a fixed informationally complete finite POVM for the single protected audit with categorical squared loss. Theorem 3.2 applies with $d = D_q^2 - 1$ in its induced probability chart, uniformly in n, M for each fixed experiment. The block length and constants may deteriorate as λ or β approach a degenerate endpoint. No common eigenbasis or free quantum observer memory is available.

Proof. The effects integrate to I, since $\int H = 0$, and satisfy $(1-\beta)I \le E_H \le (1+\beta)I$. Equation (29) is a completely positive normalized instrument. It models an unsharp measurement followed by the chosen unitary and a depolarizing channel. Its report density $1 + \beta\operatorname{tr}(H\sigma)$ is genuinely state dependent. Informational completeness separates the density matrix; every future instrument-word expectation is affine in it. This derives the predictive quotient and positive test cone from raw experiments. The probability chart is an injective affine map; use its Euclidean Brier norm. Estimates in Hilbert–Schmidt coordinates change only fixed chart constants.

Set $K = \{\sigma \ge (1-\lambda)\tau : \operatorname{tr}\sigma = 1\}$ and $D = \{\sigma \ge (1-\lambda)\tau/2 : \operatorname{tr}\sigma = 1\}$. The raw image of every positive matrix lies in C_K, including rank-deficient input matrices. Report continuity and operator domination follow from compactness and linearity. Since every normalized $q \in K$ satisfies $q \le I$, input $x \in K$ dominates κq with, for example, $\kappa = (1-\lambda)/(2D_q)$.

For fixed interior ρ, the map $E \mapsto \sqrt{E}\rho\sqrt{E}/\operatorname{tr}(E\rho)$ from positive trace-D_q effects to positive trace-one matrices has inverse

$$B_{\rho,\sigma} = \rho^{-1/2}(\rho^{1/2}\sigma\rho^{1/2})^{1/2}\rho^{-1/2}, \qquad E = D_q B_{\rho,\sigma}^2/\operatorname{tr} B_{\rho,\sigma}^2.$$

The positive solution of $B\rho B = \sigma$ is unique, which verifies both inverse identities. On $\rho \in D$ and the compact effect ball, all relevant eigenvalue floors are positive. The inverse derivative is uniformly bounded. Thus the forward Jacobian is bounded away from zero; the final affine depolarizing/unitary map is invertible on the trace hyperplane. Its actual report density is bounded, so the acquired row has a uniform full-dimensional upper density. Noncommuting effects are included.

At $\rho = \tau$ the normalized output is $\tau + (\lambda\beta/D_q)U_a H U_a^*$. An interior effect ball therefore maps over a common neighborhood B_* of τ. The smooth inverse and compactness show that for all ρ in a sufficiently small closed ball G about τ, every action's actual row dominates a common $\epsilon\zeta$, where ζ is normalized volume on a smaller B_*. This proves local actual overlap; it is not asserted globally in one step.

Use $V(\rho) = \|\rho - \tau\|_{\mathrm{HS}}$ and $E_a = \{\|H\|_{\mathrm{HS}} \le s\}$, the same quiet report ball for all actions. The matrix square-root estimate and the likelihood lower bound give

$$V(F_a(H,\rho)) \le \lambda V(\rho) + C\lambda\beta s \quad (H \in E_a,\ \rho \in K).$$

Indeed expand numerator and denominator using $\|\sqrt{I+\beta H} - I\|_{\mathrm{op}} \le C\beta s$ and $\mathrm{tr}(E_H\rho) \ge 1 - \beta s$. Each quiet report has actual probability at least $(1-\beta)s^d$, independently of the earlier actions and reports conditional on their history. Choose s so that the drift floor is below one quarter of the radius of G, choose k so that $\lambda^k \sup_K V$ is below one quarter of it, and then choose s_0 so that accumulated grid error is below one quarter of it. The fixed chart equivalence supplies the L_V in (11). All certificates are now uniform over arbitrary action rows and all sufficiently fine common grids. The good reports are paid acquisitions; the proof has not postselected a new physical preparation. Lemma 5.2 gives the joint minorization and finite-acquisition conclusion. No infinite-dimensional operator closure is used. □

8. An explicit two-state filtering benchmark

Proposition 8.1 (A two-label stationary nonlinear filter)**.** *Consider a binary hidden state, transition matrix* $P = \left(\begin{smallmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{smallmatrix}\right)$*, reports* $N(1,1)$ *and* $N(-1,1)$ *before that transition, and a protected binary Brier audit after it. Let* Φ *be the standard normal distribution function and define on* $[1/2, 3/4]$

$$G(q) = \frac{1}{4} + \frac{1}{2}\{q\Phi(1+\ell_q) + (1-q)\Phi(1-\ell_q)\}, \qquad \ell_q = \tfrac{1}{2}\log\frac{q}{1-q}. \tag{30}$$

There is a unique solution $q_* = G(q_*)$*. The homogeneous two-label observer uses programmed probabilities* $q_*, 1-q_*$*, sends a report to the high label exactly when* $y \ge -\frac{1}{2}\log(q_s/(1-q_s))$*, and outputs* $(q_s, 1-q_s)$*. Its invariant label masses are* $1/2, 1/2$ *and its exact stationary audit risk is*

$$R_\infty = 2q_*(1-q_*). \tag{31}$$

Every local common-grid observer for this experiment, with any number of labels, has a joint-ensemble minorization with $b = 1$*,* $\gamma \ge 1/120$*. In particular, the displayed two-label observer has finite-average risk at most* $2q_*(1-q_*) + 480/n$ *from every admissible initial probability and either initial label. The constants are deliberately conservative. This proposition does not assert global two-label optimality.*

Proof. Write p for the probability of hidden state one. The actual update is

$$F(y,p) = \tfrac{1}{4} + \tfrac{1}{2}\,\frac{pe^{2y}}{1-p+pe^{2y}}, \qquad K = [1/4, 3/4], \quad D = [1/8, 7/8].$$

The two cells are cut at $1/2$. A high report from programmed input q is exactly $y \ge -\ell_q$. Reflection of both the hidden state and the report exchanges the two incoming labels and the two outgoing cells. With incoming masses $1/2, 1/2$ and means $q, 1-q$, the outgoing high mass is therefore $1/2$. Its unnormalized state-one contribution is the sum of the transition-prediction terms on the two high-report events. Conditional Gaussian tail integration yields its normalized probability

$$\tfrac{1}{4} + \tfrac{1}{2}\{q\Phi(1+\ell_q) + (1-q)\Phi(1-\ell_q)\} = G(q).$$

Thus a fixed point of (30) is exactly the invariant calibration equation, not a pathwise approximate-posterior equation.

The identity $q\phi(1+\ell_q) = (1-q)\phi(1-\ell_q)$ cancels the derivative terms involving ℓ_q'. Hence

$$G'(q) = \tfrac{1}{2}\{\Phi(1+\ell_q) - \Phi(1-\ell_q)\} \in [0, 1/2].$$

Also $G(1/2) > 1/2$, $G(3/4) < 3/4$ and G is increasing. It maps the closed interval into itself and is a strict contraction. This proves existence, uniqueness and geometric computation of q_*. Conditional Brier projection at its calibrated invariant ensemble gives (31). Certified iteration, rather than

decimal approximation, defines the program constant; finite readout and comparison errors are treated in Section 11.

For the minorization, let $x \in [2/5, 3/5]$ be an acquired output and put $w = 2x - 1/2 \in [3/10, 7/10]$. For any programmed $q \in K$, the inverse report is $y = \frac{1}{2}(\log(w/(1-w)) - \log(q/(1-q)))$, so $|y| < 1$. Its actual mixture density is at least $\phi(2)$, while $dy/dx = 1/[w(1-w)] \geq 4$. The acquired scalar density is therefore at least $4\phi(2) > 1/8$ on this interval. With ζ uniform there, whose density is 5, one may take $\epsilon = 1/40$. For binary probabilities in K, the positive-cone comparison holds with $\kappa = 1/3$. Lemma 5.1 gives $\gamma \geq 1/120$ and $b = 1$ without any erasure. Binary Brier loss is at most 2, so (18) costs at most $480/n$.

For reference, the acquired density is uniformly bounded on all of K, not only on the overlap interval. If $z = \log(w/(1-w))$, its actual density is a Gaussian mixture with variance 4 and means bounded in absolute value by $B = \log 7 + 2 < 4$, uniformly on D. Its density is at most $(2\sqrt{2\pi})^{-1} \exp[-((|z| - B)_+)^2/8]$. Since $dz/dx \leq 8e^{|z|}$, the acquired scalar density is at most $4e^{B+2}/\sqrt{2\pi} < 1024$. This gives an explicit full-support density certificate; no Gaussian tail was removed. The induced Brier chart changes only the stated fixed norm constants. □

9. Weighted acquired strata

Theorem 9.1 (Finite-acquisition singular profile)**.** *Replace the full-dimensional row density by a finite visible decomposition: at each call the report mark j has fixed actual probability $w_j > 0$, independent of the incoming state and action. Given that mark, the acquired state lies in a compact convex affine set $K_j \subset K$ of dimension $d_j \geq 1$, with relative density at most Λ_j uniformly on D. Finitely many separately marked atom states are also allowed. Require the same raw interior image and robust accessibility, with the local overlapping measure now a probability on the marked acquired space. Relative grid boundaries have zero conditional mass. There is at least one positive-dimensional component.*

For all M above a fixed threshold, put

$$\Xi(M) = \sum_{j : d_j > 0} w_j M^{-2/d_j}. \tag{32}$$

Then

$$\mathcal{R}_{\mathrm{cp}}(n, M) - J_*(n) \asymp \Xi(M), \qquad c\Xi(M) \leq \mathcal{R}_{\mathrm{hom}}(n, M) - J_*(n) \leq C\big(\Xi(M) + n^{-1}\big). \tag{33}$$

One calibrated homogeneous observer attains the upper for every n, has an M-independent joint mixing bound, and has stationary excess comparable to $\Xi(M)$. Thus finite-acquisition matching holds for $n \geq c_* \Xi(M)^{-1}$. Constants can depend on the fixed positive weights, affine components and accessibility certificates. No optimal allocation across varying weights is claimed.

Proof. Projection of the marked overlap to its predictive coordinate gives the actual overlap needed for the Bellman bias. Reserve one label per atom and distribute the remaining labels in fixed positive proportions among the finitely many positive-dimensional marked grids. For sufficiently large M their maximal ambient diameter is below s_0. In the calibration correspondence use, for each label, the cone over its own compact relative cell closure; an atom has its singleton cone. Every outgoing raw contribution is supported in that cell with its actual mark. Zero-row and action-tie arguments are unchanged, and relative boundaries are null.

The cone lift integrates $(1, x)$ against the marked subprobability witness and allocates it to the matching marked label. It is positive in C_K regardless of the dimension of the cell. The robust quiet-report argument uses only ambient mesh error. It therefore gives the same budget-independent block minorization and attraction, without an erasure label. At stationarity apply Lemma 4.3 in each positive-dimensional affine span to the restriction of h, whose Lipschitz constant is inherited from

K. Each atom contributes zero cell gap. The total Jensen gap is at most $C\sum_j w_j m_j^{-2/d_j} \le C\Xi(M)$, and entrance costs C/n.

For the lower, orthogonally project any M global readouts into each component's affine span. Projection cannot increase distance. The conditional actual-density small-ball argument gives $c_j M^{-2/d_j}$. Multiply by the original w_j and sum. The same conditional projection at the actual audit cut gives the lower relative to $J_*(n)$ for every acquisition policy and every observer, without a bias endpoint subtraction. Marked grids give the checkpoint upper. Weights have not been normalized away; overlapping affine spans and off-stratum readouts cause no difficulty. □

Corollary 9.2 (Raw changing acquired rank)**.** *After the quantum instrument in Theorem 7.2, independently choose a visible mark j with probability w_j and apply a trace-preserving unital conditional expectation $\mathcal{E}_j$ onto a matrix subalgebra. Include the identity with positive weight. The completely depolarizing map may be an atom mark. Then Theorem 9.1 holds with d_j the affine dimension of the trace-one part of the subalgebra.*

Proof. The raw marked instrument is $w_j\mathcal{E}_j L_a(H)$. It is positive and normalized after summing marks. Conditional expectations fix τ and preserve the positive spectral floor, so the entire physical cone is still mapped to C_K. In Hilbert–Schmidt coordinates each expectation is an orthogonal projection. Projecting the full-dimensional bounded acquired density gives a bounded relative density by Fubini, with a bound equal to the original density constant times a maximal compact-fiber volume. The fixed audit chart only changes Jacobian constants. Depolarization gives exactly the atom τ.

For the joint accessibility witness restrict the k quiet reports and the last overlapping report to the identity mark. Their real subprobability mass is multiplied by w_{id}^{k+1}. The local overlap is supported on the identity mark, and the quiet event probability at each step is at least $w_{\mathrm{id}}(1-\beta)s^d$. All other marks and their original weights still execute on the complement. No postselection changes the experiment. Relative grids give the remaining conditions.

The acquired predictive rank changes, for example, between the full density body and a diagonal body. Physical density matrices remain strictly positive because of the stated channel. This is not an assertion of physical eigenvalue-rank change or an extension to arbitrary pure-state boundaries; the separate pure-rank theorem retains its own hypotheses in the Bayesian companion. □

10. Common-task errors and causal resources

Proposition 10.1 (Uniform perturbation of an executed observer)**.** *Let P be a common-grid observer map on C_K^m satisfying* (16)*. Let $\widetilde{P}$ be another positive mass-preserving ensemble map, with*

$$\sup_{Z\in C_K^m,\ e_{\mathrm{tot}}(Z)=1} \|(\widetilde{P}-P)Z\|_{\oplus K} \le \tau. \tag{34}$$

Writing z^* for the invariant ensemble of P, every invariant normalized ensemble $\widetilde{z}$ of $\widetilde{P}$ obeys

$$\|\widetilde{z}-z^*\|_{\oplus K} \le b\tau/\gamma. \tag{35}$$

From the same initial ensemble in C_K^m, the distance between their time-t ensembles is at most $b\tau/\gamma$ for every $t \ge 0$. The latter statement also holds for time-dependent perturbed maps having the same per-call bound. If $b\tau < \gamma$, the homogeneous $\widetilde{P}$ itself has a unique invariant ensemble and block contraction coefficient at most $1-\gamma+b\tau$.

For fixed bounded audit readouts, the stationary and finite-average risk perturbations are at most $L_0 b\tau/\gamma + e_{\mathrm{out}}$, where e_{out} is a separately certified expected loss change in the readout. The coefficient b/γ does not depend on M.

Proof. For zero-mass differences, P^j contracts by at most $(1-\gamma)^{\lfloor j/b\rfloor}$. Along a perturbed execution the one-call residual has mass zero and norm at most τ. Expanding the difference by inserting one

residual at a time therefore gives

$$\|\widetilde{Z}_t - Z_t\|_{\oplus K} \le \tau \sum_{j=0}^{t-1} (1-\gamma)^{\lfloor j/b \rfloor} \le b\tau/\gamma.$$

This uses no stationary calibration of the perturbed execution. An invariant ensemble exists by compactness and continuity of its finite-dimensional positive map. Start the comparison there and let $t \to \infty$ to prove (35).

For any difference D of total mass zero, a positive/negative base-norm decomposition and (34) give $\|(\widetilde{P} - P)D\| \le \tau\|D\|$. Both maps are contractions on their full base-norm spaces. Telescoping the bth powers bounds their operator difference on zero-mass vectors by $b\tau$. Hence $\widetilde{P}^b$ contracts that space by at most $1 - \gamma + b\tau$. Its fixed point is unique and is fixed by $\widetilde{P}$, as before. The loss is a bounded affine functional on normalized ensembles, proving the risk statement. □

Proposition 10.2 (Typed causal simulation)**.** *Fix the same protected stopping request, response space and loss in $[0, L]$. A parameter-independent causal simulator from a source experiment to the target must provide a legal strategy lift and action map, causal report transformation, R counted internal labels, a call and physical-time map, and compatible task/audit readouts. If its complete stopped and scored law differs by at most ε in total variation, uniformly over the target policies compared, an M-label target observer is implemented with at most RM source labels and risk at most its target risk plus $L\varepsilon + e_{\mathrm{out}}$.*

Compatible serial simulators multiply label budgets, compose call maps and add complete-law errors. A reverse lower requires a reverse simulator with its own resource map. A per-call full-law error alone gives a finite-prefix bound proportional to the number of calls; the uniform ensemble conclusion of Proposition 10.1 requires its stronger common-ensemble hypotheses.

Proof. Store the simulator and observer labels as a pair. Route actions through the legal lift and generate each target report without using its future or an unknown true parameter. The observer's fresh private coins are included in the complete joint law. Compatibility with the external request fixes the same scored task. For total variation defined by a supremum of event differences, integrating a loss in $[0, L]$ changes expectation by at most $L\varepsilon$. Add the separately certified readout perturbation. Causal composition and the triangle inequality prove the serial claim. This standard coupling argument is not an assertion of infinite-transcript total-variation convergence, LDP equivalence or optimality of the product state count. □

Corollary 10.3 (An attained joint resource–risk law)**.** *There is a physical extension of every experiment in Theorem 3.2, with one summed-square protected task, an unobservable calibration alternative $\pm\delta$ and an attained bit-report deficiency $\varepsilon \in [0, 1/2]$, such that for all sufficiently large M,*

$$\begin{aligned} cM^{-2/d} + \delta^2 + 2\varepsilon &\le \mathcal{R}^{\mathrm{ext}}_{\mathrm{hom}}(n, M) - J_*(n) \\ &\le C\big(M^{-2/d} + n^{-1}\big) + \delta^2 + 2\varepsilon. \end{aligned} \tag{36}$$

The pathwise maximum physical-call budget is $N = n + 3$. On the matching region $n \ge c_* M^{2/d}$, this gives $\mathcal{R}^{\mathrm{ext}}_{\mathrm{hom}} - J_*(N - 3) \asymp M^{-2/d} + \delta^2 + \varepsilon$. The weighted profile $\Xi(M)$ may replace $M^{-2/d}$ under Theorem 9.1. Numerical or model error ceilings are not automatically lower bounds.

Proof. When the hidden audit request arrives, stop all core controls and hold the core state unchanged in the explicitly enlarged post-request interface. Only then prepare an independent fair bit $B \in \{-1, 1\}$ and make one paid report call which reveals it with probability $1 - a$ and otherwise returns a marked erasure. Commit the response before a final audit reveals the core audit and B. Bit responses lie in $[-1, 1]$.

On the two prescribed bit-preparation parameters, the deficiency to an exact bit report completed before the audit is exactly $a/2$. Filling an erasure with a fair bit attains the upper. For any simulator the sum of the two error probabilities on an erased report is at least a, proving the lower. The fair-bit Bayes squared risk is exactly a, obtained with readouts $-1, 0, 1$. Set $a = 2\varepsilon$.

An independent calibration coordinate is fixed at $+\delta$ or $-\delta$ and invisible in all pre-audit reports. Averaging these two alternatives gives minimax squared risk at least δ^2, attained by output zero. Add this coordinate, the bit loss and the core loss. Since the bit is prepared after controls stop, it cannot have been stored by earlier control actions in the physical system. Its report conveys no core information conditional on the previous observer label. Arbitrary randomization and the extra terminal labels therefore do not evade the core finite-n lower. Independence and the product prior give the additive lower in (36).

For the upper use m running core labels and three post-request readout copies for each, storing the bit posterior $-1, 0, 1$. At most $4m + 2$ labels implement all roles. Taking m a fixed fraction of M preserves the geometric order and, because the core mixing constant is independent of m, also preserves the C/n entrance bound. This is a sufficient implementation, not a minimal-state claim for the enlarged alphabet. Preparation, at most n core reports, the bit report and the single joint audit total at most $n + 3$ calls. The physical post-request hold and audit timing are part of this constructed extension, not assumed freely available in an unrelated experiment. □

11. Finite precision without a shrinking mixing gap

Proposition 11.1 (Effective finite-state realization)**.** *In addition to Theorem 3.2, assume effective compact nets in every relative cell closure, computable instrument integrals, effective certificates for b, γ, s_0, and uniform effective boundary/integration moduli for the chosen grid. For $M \geq M_0$ there is a finite search over actual M-label programs, with within-cell codepoints in finite σ-nets and dyadic action rows of denominator Q, returning an observer with*

$$R_\infty \leq \rho + CM^{-2/d} + \frac{L_0 b}{\gamma}\tau_{\mathrm{tab}} + \eta + \delta_{\mathrm{out}}^2. \tag{37}$$

Here τ_{tab} is a certified uniform ensemble-map approximation error, η is the finite evaluation/search slack, and δ_{out}^2 is a certified weighted squared readout error. Every enumerated exact common-grid table has the same b, γ mixing certificate, without erasure. A legal finite-arithmetic execution with per-call ensemble error τ_{run} adds at most $L_0 b\tau_{\mathrm{run}}/\gamma$ to its finite-average risk; if homogeneous and $b\tau_{\mathrm{run}} < \gamma$, it also has an attracting invariant ensemble. The same program uses at most M retained labels.

Proof. Uniform interiority of K in B gives both $z \leq_C C_B(1, q)$ for normalized physical z and $q \in K$, and a two-sided bound

$$-C_B\|q - q'\|(1, q) \leq_C (0, q' - q) \leq_C C_B\|q - q'\|(1, q).$$

After applying the positive instrument and normalizing, these imply a report-uniform Lipschitz bound for $F_a(y, q)$ in $q \in K$. They also dominate the physical report law at any normalized input by a fixed multiple of its report law at the programmed representative. A representative error σ changes a grid decision only within a strip of width $C\sigma$ of a boundary. The actual density bound and the supplied boundary modulus bound the physical mass of that strip. On a fixed m-cell grid a computable bound of the form

$$\tau_{\mathrm{tab}} \leq C_m\sigma + C|A|/Q$$

is available. Both constants include norm equivalence with the C_K base norm. To see that this norm causes no lost factor depending on m, each misrouted positive physical contribution is in C_K; routing it to two different labels costs at most twice its mass in the sum base norm. Action-row rounding obeys the same estimate. More general supplied strip moduli can be used instead.

Choose each approximating representative in the closure of its own fixed cell. All update tables then remain local at the original mesh, not at an accumulated numerical-history radius. Enumerate these nets and all dyadic action rows. Lemma 5.2 applies to every candidate, whether or not its actions minimize the Bellman equation. The family contains a table approximating the calibrated fixed-point observer. Proposition 10.1, initially with that observer's fixed readouts, bounds its invariant risk increase by $L_0 b\tau_{\mathrm{tab}}/\gamma$.

For each finite candidate, compute its actual invariant ensemble by certified finite instrument integration and positive iteration. The truncation error after t calls is at most $2(1-\gamma)^{\lfloor t/b \rfloor}$; integration errors are amplified by at most b/γ. Recenter the readouts at the conditional means of this actual candidate ensemble. This cannot increase risk. Very small-mass labels need not be divided by an interval containing zero: when their total upper mass is below an assigned squared-error allowance divided by L_0, use arbitrary legal readouts there. Refine all other mass intervals until positive lower bounds allow division. Because there are finitely many labels, the positive masses are eventually resolved and the zero masses can be charged by their shrinking upper bounds. Approximate the remaining means to achieve total weighted squared error δ_{out}^2. Finite interval evaluation of all candidates and further refinement supplies slack η and proves (37).

A running finite-arithmetic interpreter still sees the original physical report kernel. Approximate comparisons, tails and fallbacks are bounded by a certified uniform per-call ensemble discrepancy. They do not redefine the experiment by conditioning on a bounded report region. Proposition 10.1 supplies the uniform risk estimate; its block contraction argument gives attraction when the error is small enough. A homogeneous implementation must not retain a growing precision counter or an uncounted random seed. □

The search size is at most

$$\prod_{s=1}^{m} |\operatorname{Net}(\overline{C}_s, \sigma)|\,(Q+1)^{m|A|}$$

before finite readout refinement. No efficient global synthesis claim is made. To retain the geometric risk order it is sufficient to take $\tau_{\mathrm{tab}}+\tau_{\mathrm{run}} \lesssim M^{-2/d}$ and $\eta+\delta_{\mathrm{out}}^2 \lesssim M^{-2/d}$, with constants depending on b/γ. Unlike a vanishing-erasure construction, this does not require an additional factor $M^{-2/d}$ in the ensemble error tolerance.

The persistent label uses $\lceil \log_2 M \rceil$ bits. A direct table description has upper size

$$O\big(Md\log(1/\sigma) + M|A|\log Q + Md\,p_{\mathrm{out}}\big) + \text{shared primitive and grid description.}$$

Its workspace is bounded by $W_{\mathrm{primitive}}(p) + O(dp + \log M + |A|\log Q)$ for a declared primitive precision p; time includes primitive evaluation, cell search and fresh coin generation. Offline enumeration and integration have their own costs. The readout precision p_{out} must be certified using actual label masses; it is not automatically a universal function of δ_{out}. The external audit clock and simulator states are separate resources. Effective Gaussian tail/log-odds and compact quantum spectral-floor certificates provide the stated integration and strip assumptions when their model parameters are effectively given. Arbitrary real kernels are not presumed to have finite descriptions.

12. Comparison, consequences and scope

Sequential quantization has long treated causal encoders rather than independent static codebooks; Borkar, Mitter and Tatikonda [1] supply a central comparison. The present observer also chooses the acquisition action. Its retained labels are the entire online information available for both control and prediction. A belief-grid approximation accompanied by a separately maintained exact filter would not have this resource interpretation.

The existence of near-optimal finite-state controllers is not new. Yu and Bertsekas [13] establish such an average-cost approximation result for a finite POMDP under their constant-optimal-average-cost hypothesis. General finite approximations of Borel MDPs are treated by Saldi, Yüksel and Linder [11]. Those results motivate, but do not themselves give, the joint acquired-density lower, the total-label rate, the continuous-report calibrated implementation and the budget-independent entrance and perturbation certificates proved here. Our synthesis proof is a finite exhaustive approximation argument, not a computational improvement on those theories.

For partially observed average control, Demirci, Kara and Yüksel [2] derive average-cost optimality and approximation conclusions through nonlinear-filter contraction. We use a standard span-contraction Bellman argument after proving its feedback-uniform acquired-overlap premise. The step not supplied by that Bellman equation is the invariant calibration and mixing of the executed finite observer. A filter can be stable while an observer's memory stays trapped in incompatible classes. Lemma 5.1 concerns the joint finite-label instrument, not just the full-history filter.

Value-directed compression [7] focuses on preserving the decision-relevant geometry of a POMDP. The Bayes-risk/Bregman connection [12] explains why curvature is naturally task dependent. Here the protected squared task makes conditional means exact and gives a uniform strong-concavity lower. The upper uses a nonsmooth Bellman bias and actual mass per grid cell. We do not assert that a different loss has the same exponent without an appropriate replacement curvature certificate.

The finite-memory belief approximation analysis of Kim [5] controls errors using a contracting compressor and fixes the policy and closed-loop execution in its comparison. We distinguish that comparison from the infimum over all adaptive acquisition policies in $J_*(n)$. We do not claim that an arbitrary compressor is contractive. The common-grid implementation in this paper receives a separate positive-cone proof of joint mixing, before its optimal stationary calibration is selected.

The general hypotheses remain sufficient, not necessary. Whole-base interiority is used exactly when the remainder in a positive-cone comparison is passed through a raw instrument. Local acquired overlap without robust accessibility does not give our uniform observer minorization. Common small-ball mass without a global legal recursive cover does not supply an implementable upper. Section A records these distinctions. The two raw realizations do not reverse these implications.

For the stated class, the finite-acquisition conclusion improves the erasure-dependent entrance term and precision tolerance of the retained stationary article [10]: there is no vanishing erasure rate and no extra label lane. The conditional geometry remains that of the actually executed law. The singular result retains acquired mixture weights and permits the support dimension to change at every call. It does not remove every degeneracy: arbitrary nondominated kernels, infinite-dimensional predictive closure, unknown-kernel learning, nonsummable inaccessible classes, arbitrary singular stratifications, optimal program/workspace/time tradeoffs and unrestricted unbounded stopping require additional theorems. No general impossibility claim is made about those tasks.

Appendix A. Interfaces and distinctions used in the proof

The predictive-quotient assertion assumes a finite separating test realization and a Borel realization section. It does not infer either from a formal coordinate count. Positivity gives the affine unnormalized instrument, and likelihood division is made only off a common zero set. The Gaussian and quantum constructions explicitly verify that the protected audit separates the physical state, so these are predictive coordinates rather than an arbitrary enlargement.

Local calibration is not global calibration. For example, let two incoming rows be point masses at distinct values x_0, x_1 and let each use a single output label. Each row's one-center quantization error is zero, but merging the two rows into the same label has strictly positive conditional variance whenever both rows have positive mass. Even vanishing pooling error at one update does not

prove that the centroids repeat at the next update. Lemma 5.3 solves the entire invariant ensemble equation, and Lemma 5.2 proves attraction for the resulting fixed program.

Mixing of the physical instrument alone is insufficient for mixing of an arbitrary observer. A program that never changes its incoming label has several invariant label classes, even when the physical kernel is independent fresh sampling at each call. The common-grid locality hypothesis rules out this construction by forcing labels to name cells of their programmed acquired response. The positive-cone remainder in Lemma 5.1 lies initially in C, not necessarily in C_K; positivity only on a restricted normalized domain would not justify its image. This is why (6) is stated on the entire physical cone.

The joint minorization describes ensembles of executable test responses. It does not identify their base norm with total variation on an unrepresented microscopic path space. Proposition 10.2 therefore assumes a separate complete stopped-law deficiency when transferring between experiments. Conversely, finite-arithmetic report rounding is not claimed to be close in total variation to a continuous report law; the relevant bound is on the instrument after its label decision and on the scored readout.

A homogeneous observer uses one table at every call. The uniform audit request is generated by an independent paid external mechanism and is not an internally exact deadline. A visible end marker, a timer supplied to the observer or a readout depending on an uncounted phase would change the legal resource model. The preparation and final audit are physical calls. The finite program, its work memory and computation are not measured by $\log_2 M$ alone.

Finally, the mass density and lower-dimensional component laws used for lower bounds are actual acquired probabilities. Rare report paths in the accessibility proof remain subprobabilities of their original mass. They are not postselected free preparations. A similarity between two formal supports gives none of these mass, recurrence or implementation statements.

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